SIAM J. APPL. MATH.
Vol. 32, No. 2, March 1977

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

ON THE OPTIMALITY OF STRUCTURED POLICIES IN COUNTABLE STAGE DECISION PROCESSES. II: POSITIVE AND NEGATIVE PROBLEMS*

DAVID M. KREPS AND EVAN L. PORTEUS†

Abstract. The analysis of structured countable stage decision processes, initiated in Porteus [11], is continued. The standard models of positive and negative dynamic programming are given in this context, thus extending these results to criteria other than the usual expected sum of rewards, such as expected utility criteria, certain stochastic games, risk sensitive Markov decision processes, and maximin criteria.

For positive problems, (what are called) unimprovable strategies are optimal and the optimal value sequence is the least solution of the optimality equations exceeding an obvious lower bound. For negative problems, conserving strategies are optimal, and if one strategy is a one-step improvement on another, then it nets a greater value. (This rules out cycling in the strategy iteration procedure.) Also, transfinite methods are used to prove that the optimal value sequence is the greatest solution of the optimality equations less than an obvious upper bound. We indicate how all these results can be extended to analogues of the essentially positive, essentially negative, and convergent cases.

Introduction and summary. This paper continues the study of structured countable stage decision processes, initiated in [11]. The general approach includes as a subcase the standard criterion of maximizing the expected (perhaps discounted) sum of rewards (e.g., see [1], [2], [4], [5], [9], [13]). It also encompasses expected utility criteria (see [7]) and certain other criteria, such as certain stochastic games (see [4]), risk sensitive Markov decision problems (see [11]), and maximin criteria. The reader is assumed to be familiar with the concepts and methods employed in [11], although a very brief summary of pertinent definitions and results is given in § 1.

In §§ 2 and 3, we extend the standard theories of positive and negative dynamic programming to this framework. The results obtained for positive problems are straightforward extensions of results of Blackwell [2], Hinderer [5] and Hordijk [6]. For example, under appropriate assumptions, (what we call) unimprovable strategies are optimal and the optimal value functions are the least solution of the optimality equations exceeding an obvious lower bound. For negative problems (cf. Hinderer [5], Hordijk [6] and Strauch [13]), we obtain the standard results that conserving strategies are optimal and that if one strategy is a one-step improvement on another, then it nets a higher value. (This would be needed in a strategy iteration procedure, to rule out cycling.) Using transfinite methods, we show (again with appropriate assumptions) that the optimal value functions are the greatest solution of the optimality equations less than an obvious upper bound. As in [11], we also focus on structure, e.g., we establish conditions under which the optimal value functions are structured.

A characteristic of the approach taken here is that infinite stage problems are defined indirectly, as limits of finite stage problems. In § 4, the notion of *equivalent terminal value sequence* is given: these are terminal value sequences which yield

* Received by the editors August 8, 1975.

† Graduate School of Business, Stanford University, Stanford, California 94305. This research was supported in part by a grant from the Atlantic Richfield Foundation to the Stanford Graduate School of Business.

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

(for a given problem) the same infinite stage value functions. This notion allows us to extend the analyses of §§ 2 and 3 to the cases of *essentially positive*, *essentially negative*, and *convergent* problems.

In § 5, the approach taken here (and in [11]) is contrasted with the standard theories of dynamic programming. In particular, the notion of an infinite stage problem and the consequences of our assumption (PAε) (see § 1) are discussed.

In the interest of brevity, details and examples are often only sketched or are omitted altogether. Some of these are compiled in the notes accompanying [8].

1. General definitions and results. The basic objects of a (nonstationary, denumerable stage) decision process are the *state spaces* $\{\Omega_t\}$ (for each $t=1,2,\cdots$), *decision sets* $\{D_t(x)\}$ for each t and $x\in\Omega_t$, *one transition value functions* $\{h_t\}$, and the sequence of *terminal value functions* $\{u_t\}$. Also given are spaces of *admissible value functions* $\{V_t\}$, *structured value functions* $\{V_t^*\}$, *admissible policies* $\{\Delta_t\}$, and *structured policies* $\{\Delta_t^*\}$. Constructed from all these are the *strategy space* Π, the *structured strategy space* Π^*, *single stage transition operators* $H_{t\delta}$ (for $\delta\in\Delta_t$, alternatively denoted $H_t(\pi)$ for $\pi\in\Pi$) and A_t, *multistage operators* $H_t^n(\pi)$ and A_t^n, and *finite horizon value* and *optimal value functions*, $v_t^n(\pi)$ and f_t^n. Also constructed are the *infinite horizon value* and *optimal value functions*, according to the following definitions: $v_t^*(\pi)\equiv\limsup_n v_t^n(\pi)$; $v_t(\pi)\equiv\lim_n v_t^n(\pi)$ when it exists; $f_t\equiv\sup_\pi v_t^*(\pi)$. (All limits are taken pointwise. For $v_t(\pi)$ to be defined, the limit must exist for all $x\in\Omega_t$.)

A strategy π is said to be *ε-optimal* (for $\varepsilon\geqq 0$) if, for all t, $v_t(\pi)$ exists and $v_t(\pi)\geqq f_t-\varepsilon$. A 0-optimal strategy is *optimal*. A strategy π is *conserving* if $H_t(\pi)f_{t+1}=f_t$ for all t. A strategy π is *unimprovable* if, for all t, $v_t(\pi)$ exists, $A_tv_{t+1}(\pi)$ is well-defined, and $v_t(\pi)\geqq A_tv_{t+1}(\pi)$. Strategy π is a *one-step improvement* on strategy π' if $\{v_t(\pi')\}$ and $\{H_t(\pi)v_{t+1}(\pi')\}$ are well-defined and $H_t(\pi)v_{t+1}(\pi')\geqq v_t(\pi')$ for all t, with strict inequality for some t and $x\in\Omega_t$. The *optimality equations* are the equations $f_t=A_tf_{t+1}$. (There is no a priori reason to believe that the optimality equations hold. Indeed, one of the objectives of this study is to develop conditions under which they do hold. There is a subtle difference between the approach used here and the standard treatment; this is discussed in § 5.)

We invoke the following two assumptions throughout the paper.

ASSUMPTION R (Regularity). For every t, the following hold: $\varnothing\neq\Delta_t^*\subseteq\Delta_t$; for each $x\in\Omega_t$ and $y\in D_t(x)$, there exists $\delta\in\Delta_t$ such that $\delta(x)=y$; $V_t^*\subseteq V_t\subseteq\Omega_t^{[-\infty,\infty]}$, $H_{t\delta}$: $V_{t+1}\to V_t$ for all $\delta\in\Delta_t$; $H_{t\delta}$ is isotone for all $\delta\in\Delta_t$; there exists a positive real number α_t such that $H_{t\delta}$ is α_t-Lipschitzian on V_{t+1} for all $\delta\in\Delta_t$; and $u_t\in V_t^*$.

ASSUMPTION PAε (Preservation and ε-attainment). For all t, A_t: $V_{t+1}^*\to V_t^*$. For all t, $v\in V_{t+1}^*$, and $\varepsilon>0$, there exists $\delta\in\Delta_t^*$ such that $H_{t\delta}v+\varepsilon\geqq A_tv$.

We shall sometimes assume a strengthened form of (PAε), replacing the $\varepsilon>0$ there with 0. This will be referred to as (PA). The reader comparing these assumptions with those in [11] will note that (R) is the conjunction of 1*, 2* (slightly and inessentially weakened), 3*, 6* and the requirement that $u_t\in V_t^*$ for all t; (PAε) is the conjunction of 4* and 5*, and (PA) the conjunction of 4* and 5**. Consequently, we have the following abridgement of Theorem 1 from [11].

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

THEOREM 1. *If* (R) *and* (PAε) *hold, then* $f_t^n = \sup_{\pi\in\Pi^*} v_t^n(\pi) \in V_t^*$ *and* $f_t^n = A_t f_{t+1}^n$ *for all* t *and* $n > t$.

In addition, we require the following lemma, whose proof is essentially given in [11].

LEMMA 2. *If* (R) *and* (PAε) *hold, and* $\{v_t\}$ *is a sequence of structured value functions* ($v_t \in V_t^*$) *such that* $v_t = A_t v_{t+1}$ *for all* t, *then for every* $\varepsilon > 0$, *there exists* $\pi \in \Pi^*$ *such that* $H_t^n(\pi)v_{n+1} + \varepsilon \geqq A_t^n v_{n+1} = v_t$, *simultaneously for all* t *and* $n \geqq t$.

2. Positive problems. In addition to assumptions (R) and (PAε), throughout this section the following three assumptions are made:

ASSUMPTION IC (Increasing closure). For every t, if $\{v^n\}$ is a nondecreasing sequence of functions in V_t (V_t^*), then $\lim_n v^n \in V_t$ (V_t^*).

ASSUMPTION IMC* (Increasing monotone convergence*). For every t, if $\{v^n\}$ is a nondecreasing sequence of functions in V_{t+1}, then $H_{t\delta}(\lim_n v^n) = \lim_n (H_{t\delta} v^n)$ for all $\delta \in \Delta_t^*$.

ASSUMPTION P* (Positive*). For all t and $\delta \in \Delta_t^*$, $H_{t\delta} u_{t+1} \geqq u_t$.

In addition, we denote by (IMC) and (P) the assumptions (IMC*) and (P*), respectively, when the policy δ in those assumptions is allowed to be any admissible policy.

The analysis makes repeated use of the following simple fact from real analysis. A proof may be found in Hinderer [5, Lemma 3.4].

LEMMA 3. *If* X *is an arbitrary* (*index*) *set and, for each* $x \in X$, $\{r_n(x)\}$ *is a nondecreasing sequence of extended real numbers, then* $\lim_n \sup_x r_n(x) = \sup_x \lim_n r_n(x)$.

With this, our result for the positive case may be given.

THEOREM 4. *If* (R), (PAε), (IC), (IMC*) *and* (P*) *hold, then*

(a) $v_t(\pi)$ *exists and is admissible for each* $\pi \in \Pi^*$ (*and each* t),

(b) $v_t(\pi) = H_t(\pi)v_{t+1}(\pi)$ *for each* $\pi \in \Pi^*$,

(c) *if in addition* (IMC) *and* (P) *hold, then* (a) *and* (b) *are true for all* $\pi \in \Pi$,

(d) $f_t = \lim_n f_t^n = \sup_{\pi\in\Pi^*} v_t(\pi) \in V_t^*$,

(e) *the optimality equations hold*: $f_t = A_t f_{t+1}$,

(f) $\{f_t\}$ *is the least majorant* $\{g_t\}$ *of* $\{u_t\}$ *which satisfies* $g_t \geqq A_t g_{t+1}$,

(g) *if* π *is optimal, then* π *is unimprovable, and*

(h) *if* π *is unimprovable and either* $\pi \in \Pi^*$ *or* (P) *holds, then* π *is optimal.*

Proof. To get (a) and (b), fix $\pi \in \Pi^*$. By (P*) and isotonicity of $H_t^n(\pi)$, $v_t^n(\pi) = H_t^n(\pi)u_{n+1} \leqq H_t^n(\pi)(H_{n+1}(\pi)u_{n+2}) = H_t^{n+1}(\pi)u_{n+2} = v_t^{n+1}(\pi)$. Thus $v_t^n(\pi)$ is nondecreasing in n and $\lim_n v_t^n(\pi)$ (exists and) is admissible. Hence using (IMC*),

$$v_t(\pi) = \lim_n v_t^n(\pi) = \lim_n H_t(\pi)v_{t+1}^n(\pi) = H_t(\pi)\big(\lim_n v_{t+1}^n(\pi)\big) = H_t(\pi)v_{t+1}(\pi).$$

Part (c) follows similarly.

Use (R) to pick some $\pi_0 \in \Pi^*$. Then by (P*), $u_t \leqq H_t(\pi_0)u_{t+1} \leqq A_t u_{t+1} = f_t^t$, and by isotonicity, $u_t \leqq f_t^t \leqq f_t^{t+1} \leqq \cdots$. Thus $\lim_n f_t^n$ exists and is structured by (IC). Using Theorem 1, Lemma 3, and Theorem 4(a),

$$\lim_n f_t^n = \lim_n \sup_{\pi\in\Pi^*} v_t^n(\pi) = \sup_{\pi\in\Pi^*} \lim_n v_t^n(\pi) = \sup_{\pi\in\Pi^*} v_t(\pi).$$

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Since $f_t^n \geqq v_t^n(\pi)$ for all $\pi \in \Pi$,

$$\lim_n f_t^n \geqq \sup_{\pi\in\Pi} \lim_n \sup v_t^n(\pi) = \sup_{\pi\in\Pi} v_t^*(\pi) = f_t \geqq \sup_{\pi\in\Pi^*} v_t(\pi).$$

Thus $\lim_n f_t^n = f_t = \sup_{\pi\in\Pi^*} v_t(\pi)$, which is (d).

For (e), use (PAε), (IMC*), (3), and (d) to get

$$f_t = \lim_n f_t^n = \lim_n A_t f_{t+1}^n = \lim_n \sup_{\pi\in\Pi^*} H_t(\pi) f_{t+1}^n = \sup_{\pi\in\Pi^*} \lim_n H_t(\pi) f_{t+1}^n$$
$$= \sup_{\pi\in\Pi^*} H_t(\pi)[\lim_n f_{t+1}^n] = \sup_{\pi\in\Pi^*} H_t(\pi) f_{t+1} = A_t f_{t+1}.$$

For (f), suppose $\{g_t\}$ satisfies $u_t \leqq g_t$ and $g_t \geqq A_t g_{t+1}$. By isotonicity, $g_t \geqq A_t^n g_{n+1} \geqq A_t^n u_{n+1} = f_t^n$; thus $g_t \geqq \lim_n f_t^n = f_t$.

Finally, suppose π is optimal. Then $\{v_t(\pi)\}$ exists and

$$v_t(\pi) = f_t = A_t f_{t+1} = A_t v_{t+1}(\pi);$$

thus π is unimprovable. If π is unimprovable, then $\{v_t(\pi)\}$ exists and satisfies $v_t(\pi) \geqq A_t v_{t+1}(\pi)$. Either (P*) (if $\pi \in \Pi^*$) or (P) implies that $v_t(\pi) \geqq u_t$. Thus by (f), $v_t(\pi) \geqq f_t$ and π is optimal. □

From Theorem 4(d), it is (theoretically) straightforward to find the optimal value functions. However, even assuming (PA) (so that a conserving strategy exists), there will not always exist an optimal strategy. A standard counterexample is Example (3.2) of Blackwell [2].

So for positive problems, one will have to be content in general with "almost" optimal strategies. Using Theorem 4 (d), for any t, x and $\varepsilon > 0$, one can produce a (structured) strategy π such that $v_t(\pi)(x) > f_t(x) - \varepsilon$ (with the obvious modification if $f_t(x) = \infty$). In general, however, (uniformly) ε-optimal strategies may not exist. That is, given t and $\varepsilon > 0$, it may be impossible to find a strategy π such that $v_t(\pi)(x) \geqq f_t(x) - \varepsilon$ (or $v_t(\pi)(x) \geqq f_t(x)(1-\varepsilon)$) for all $x \in \Omega_t$. An example can be found in [8].

3. Negative problems. In this section, problems satisfying the following three assumptions (in addition to (R) and (PAε)) are studied.

ASSUMPTION DC (Decreasing closure). For every t, if $\{v^n\}$ is a nonincreasing sequence of functions in V_t (V_t^*), then $\lim_n v^n \in V_t$ (V_t^*).

ASSUMPTION DMC* (Decreasing monotone convergence*). For every t, if $\{v^n\}$ is a nonincreasing sequence of functions in V_{t+1}, then $H_{t\delta}[\lim_n v^n] = \lim_n H_{t\delta} v^n$ for all $\delta \in \Delta_t^*$.

ASSUMPTION N (Negative). For every t, $A_t u_{t+1} \leqq u_t$.

(DC) and (DMC*) are obvious analogues to (IC) and (IMC*). We denote by (DMC) the strengthening of (DMC*) to hold for all $\delta \in \Delta_t$. Note that, as long as (PAε) holds, (N) is equivalent to $H_{t\delta} u_{t+1} \leqq u_t$ for all $\delta \in \Delta_t^*$.

The analysis of the previous section cannot be paralleled here, because Lemma 3 fails to hold if the sequences $r_n(x)$ are nonincreasing instead of nondecreasing. Thus we begin with a partial result assuming only (R), (PAε), (DC), (DMC*) and (N). A further assumption is then made which leads to additional results.

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

THEOREM 5. *If* (R), (PAε), (DC), (DMC*) *and* (N) *hold, then*

(a) $v_t(\pi)$ *exists and is admissible for all* $\pi \in \Pi$ (*and all* t),

(b) $v_t(\pi) = H_t(\pi)v_{t+1}(\pi)$ *for all* $\pi \in \Pi^*$ (*and assuming* (DMC), *this holds for all* π),

(c) $f_t^\infty = \lim_n f_t^n$ *exists and is structured,*

(d) $f_t^\infty \geqq f_t$,

(e) $f_t^\infty \geqq A_t f_{t+1}^\infty$,

(f) *if* $f_t \in V_t^*$ *and the optimality equations hold, then* $f_t = \sup_{\pi \in \Pi^*} v_t(\pi)$ *and there exists, for each* $\varepsilon > 0$, *an* ε*-optimal structured strategy,*

(g) *if* $f_t^\infty = A_t f_{t+1}^\infty$ *for all* t, *then* $f_t = f_t^\infty$ (*and* (f) *applies*),

(h) *if* π *is a one-step improvement on* π', *then* $v_t(\pi) \geqq v_t(\pi')$ *for all* t *with strict inequality for some* t *and* $x \in \Omega_t$ *if* $\pi \in \Pi^*$ *or if* (DMC) *holds,*

(i) *every optimal structured strategy is conserving* (*and assuming* (DMC), *every optimal strategy is conserving*), *and*

(j) *every conserving strategy is optimal* (*assuming* $f_t \in V_t$, *so this makes sense*); *thus if the optimality equations hold, then the optimality criterion holds*: *if* $H_t(\pi)f_{t+1} = A_t f_{t+1} (= f_t)$, *then* π *is optimal. Hence, if the optimality equations hold,* $f_t \in V_t^*$ *and* (PA) *holds, then there exists an optimal structured strategy.*

Proof. Note that by (N), for all t and π, $v_t^t(\pi) = u_t \geqq A_t u_{t+1} \geqq H_t(\pi)u_{t+1} = v_t^{t+1}(\pi)$. Thus, by the isotonicity of $H_t(\pi)$, $\{v_t^n(\pi)\}$ is nonincreasing in n. Parts (a) and (b) then follow from (DC) and (DMC*) (or (DMC)), exactly as in the proof of Theorem 4(a, b, c).

Also, as $f_t^n = \sup_\pi v_t^n(\pi)$, $\{f_t^n\}$ is nonincreasing in n, and (c) follows from (DC) and Theorem 1. For (d), note that $f_t^\infty = \lim_n f_t^n = \lim_n \sup_\pi v_t^n(\pi) \geqq \sup_\pi \lim_n v_t^n(\pi) = \sup_\pi v_t(\pi) = f_t$.

For (e), $f_t^\infty = \lim_n f_t^n = \lim_n A_t f_{t+1}^n = \lim_n \sup_{\pi \in \Pi^*} H_t(\pi)f_{t+1}^n$ [by (PAε)] $\geqq \sup_{\pi \in \pi^*} \lim_n H_t(\pi)f_{t+1}^n = \sup_{\pi \in \pi^*} H_t(\pi)[\lim_n f_{t+1}^n]$[by DMC*] $= A_t f_{t+1}^\infty$.

For (f), suppose $f_t \in V_t^*$ and $f_t = A_t f_{t+1}$ for all t. Pick $\varepsilon > 0$. Using Lemma 2, there exists a strategy $\pi \in \Pi^*$ such that $H_t^n(\pi)f_{n+1} \geqq f_t - \varepsilon$ for all t and $n \geqq t$. Since $v_t(\tilde{\pi}) \leqq v_t^t(\tilde{\pi}) = u_t$ for all t and $\tilde{\pi}$, $u_{n+1} \geqq f_{n+1}$, thus by isotonicity, $v_t^n(\pi) = H_t^n(\pi)u_{n+1} \geqq H_t^n(\pi)f_{n+1} \geqq f_t - \varepsilon$ for all t and $n \geqq t$. Passing n to infinity and using (a) gives the result.

For (g), proceed as in (f) using Lemma 2 to obtain $\pi \in \Pi^*$ such that $H_t^n(\pi)f_{n+1}^\infty \geqq f_t^\infty - \varepsilon$ for all t and $n \geqq t$ (and $\varepsilon > 0$ fixed). Since $u_t = f_t^t \geqq f_t^\infty$, isotonicity yields $v_t^n(\pi) = H_t^n(\pi)u_{n+1} \geqq H_t^n(\pi)f_{t+1}^\infty \geqq f_t^\infty - \varepsilon \geqq f_t - \varepsilon$ [by (d)]. Passing n to infinity and using (a) then gives the result.

For (h), suppose π is a one-step improvement on π'. Then $v_t^t(\pi) = u_t \geqq v_t(\pi')$ for all t. Assume inductively on n that $v_t^{t+n}(\pi) \geqq v_t(\pi')$ for all t. Then $v_t^{t+n+1}(\pi) = H_t(\pi)v_{t+1}^{t+n+1}(\pi) \geqq$ [by the induction hypothesis] $H_t(\pi)v_{t+1}(\pi') \geqq$ [since π is a one-step improvement on π'] $v_t(\pi')$. Thus by induction $v_t^n(\pi) \geqq v_t(\pi')$ for all t and n, and $v_t(\pi) = \lim_n v_t^n(\pi) \geqq v_t(\pi')$. Furthermore, if $\pi \in \Pi^*$ or (DMC) holds, $v_t(\pi) = H_t(\pi)v_{t+1}(\pi)$. Since π is a one-step improvement on π', for some t and $x \in \Omega_t$, $H_t(\pi)v_{t+1}(\pi')(x) > v_t(\pi')(x)$. But then $v_t(\pi')(x) < H_t(\pi)v_{t+1}(\pi')(x) \leqq H_t(\pi)v_{t+1}(\pi)(x) = v_t(\pi)(x)$.

Suppose $\pi \in \Pi^*$ is optimal. Then using (b), $f_t = v_t(\pi) = H_t(\pi)v_{t+1}(\pi) = H_t(\pi)f_{t+1}$, so π is conserving. (Assuming (DMC), the proof for optimal $\pi \in \Pi$ is similar.) This is (i).

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Finally, for (j), suppose π is conserving. Then $f_t = H_t^n(\pi)f_{n+1} \leqq H_t^n(\pi)u_{n+1} = v_t^n(\pi)$ for all t and n. Passing n to infinity and using (a) gives the result. □

The primary deficiency in Theorem 5 is that we can prove neither that $f_t^\infty = f_t$ nor that $f_t = A_t f_{t+1}$. Indeed, these are not true in general. A simple counterexample to $f_t^\infty = f_t$ (given in the standard terminology of a stationary Markov decision chain with additive rewards) is given here (see Fig. 1). (A counterexample to $f_t = A_t f_{t+1}$ can be constructed from this example, using a nonexpected utility criterion; see [8].) There are countably many states $\{\omega, 0, 1, 2, \cdots\}$. State 0 is absorbing. In state n $(= 1, 2, \cdots)$, only one action can be taken and transition is to state $n-1$ with certainty. In state ω, there are countably many actions that can be taken, $\{1, 2, \cdots\}$; if action n is taken, transition is to state n with certainty.

In the transition from 1 to 0, a reward of -1 is obtained; otherwise the reward is 0. (Rewards are added without discounting.) Now starting in state ω, every strategy π will net -1 eventually: $v_t(\pi)(\omega) = -1$ for all π, thus $f_t(\omega) = -1$. But, if the finite horizon returns are taken as just the partial sum of rewards received up to the horizon (which is typical), for every n we can find a strategy π (namely, on the first transition, go to $n+1$ starting in ω) such that $v_t^n(\pi)(\omega) = 0$. Thus $f_t^n(\omega) = 0$ for all n and $f_t^\infty(\omega) = 0$. What is happening here should be obvious: starting in ω, one can delay the "bad" transition from 1 to 0 for any finite amount of time, but not (using an admissible strategy) forever.

To get further results, we require the following strengthened form of (DC). (Also, (DMC) will be required.)

ASSUMPTION GDC (General decreasing closure). V_t is closed under nonincreasing sequences (just as in (DC)) and V_t^* is closed under nonincreasing transfinite sequences. That is, if $(\mathscr{B}, \succcurlyeq)$ is a linearly ordered set and if, for each $\beta \in \mathscr{B}$, $\{g_t^\beta\}$ is a sequence of functions such that (i) $g_t^\beta \in V_t^*$ for all t and β and (ii) $g_t^\beta \leqq g_t^\alpha$ if $\beta \succcurlyeq \alpha$, then $\inf_{\beta \in \mathscr{B}} g_t^\beta \in V_t^*$.

This assumption will be discussed in the sequel. But first, we provide the main result and a corollary. Although it may not be apparent in the proof of the theorem, there is a good deal of intuitive content in this line of analysis, which is provided in the discussion preceding the corollary.

THEOREM 6. *If* (R), (PAε), (GDC), (DMC) *and* (N) *hold, then* $f_t \in V_t^*$ *and* $f_t = A_t f_{t+1}$ *for every* t.

Proof. Let U be the collection of all sequences of functions $\{\bar{u}_t\}$ such that, for all t, (i) $\bar{u}_t \in V_t^*$, (ii) $u_t \geqq \bar{u}_t \geqq f_t$, and (iii) $\bar{u}_t \geqq A_t \bar{u}_{t+1}$. This set is nonempty: $\{u_t\} \in U$. Partially order U by the rule: $\{\bar{u}_t\} \succcurlyeq \{\bar{u}_t'\}$ if $\bar{u}_t \leqq \bar{u}_t'$ for all t.

For each $\{\bar{u}_t\} \in U$, a problem can be defined using $\{\bar{u}_t\}$ as the terminal reward functions in place of $\{u_t\}$. Let $\bar{v}_t^n(\pi)$, $\bar{v}_t(\pi)$, etc., denote the resulting value

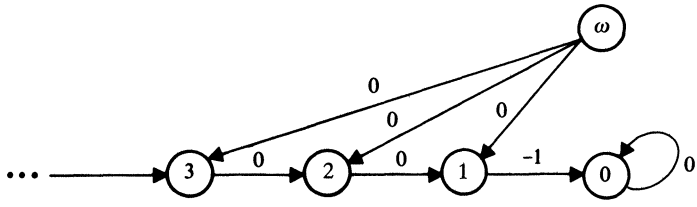

FIG. 1. *Pictorial representation of example*

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

functions. By (iii), this new problem satisfies (N) (of course, the other assumptions remain in force) so Theorem 5 applies. Also, for every π, $v_t(\pi) = \bar{v}_t(\pi)$, since by (ii), Theorem 5(a, b) and isotonicity,

$$v_t(\pi) = \lim_n H_t^n(\pi)u_{n+1} \geqq \bar{v}_t(\pi) = \lim_n H_t^n(\pi)\bar{u}_{n+1}$$
$$\geqq \lim_n H_t^n(\pi)f_{n+1} \geqq \lim_n H_t^n(\pi)v_{n+1}(\pi) = v_t(\pi).$$

Thus, $f_t = \bar{f}_t$ (for all t).

If $\{\{\bar{u}_t^\beta\}, \beta \in \mathscr{B}\}$ is a chain in $\{U, \succcurlyeq\}$, then the chain has an upper bound in $\{U, \succcurlyeq\}$, namely $\{\inf_\beta \bar{u}_t^\beta\}$. From (GDC), it is immediate that $\inf_\beta \bar{u}_t^\beta \in V_t^*$ for all t, that $u_t \geqq \inf_\beta \bar{u}_t^\beta \geqq f_t$ is apparent, and that (iii) holds, as

$$A_t[\inf_\beta \bar{u}_{t+1}^\beta] = \sup_\pi H_t(\pi)[\inf_\beta \bar{u}_{t+1}^\beta] \leqq \sup_\pi \inf_\beta H_t(\pi)\bar{u}_{t+1}^\beta$$
$$\leqq \inf_\beta \sup_\pi H_t(\pi)\bar{u}_{t+1}^\beta = \inf_\beta A_t\bar{u}_{t+1}^\beta \leqq \inf_\beta \bar{u}_t^\beta.$$

Thus Zorn's lemma can be applied, yielding a $\succcurlyeq$-maximal element of U; denote one such by $\{\tilde{u}_t\}$.

We show that $\tilde{u}_t = A_t\tilde{u}_{t+1}$. In view of (iii), it suffices to assume that for some t and $x \in \Omega_t$, $\tilde{u}_t(x) > A_t\tilde{u}_{t+1}(x)$ and to thereby derive a contradiction. Since $\{\tilde{u}_t\} \in U$, a negative problem (with value functions $\tilde{v}_t^n(\pi)$, etc.) can be defined using $\{\tilde{u}_t\}$ as the terminal value functions. But consider the resulting $\{\tilde{f}_t^\infty\}$; it is evident that $\{\tilde{f}_t^\infty\} \in U$ and that $\tilde{f}_t^\infty \leqq \tilde{u}_t$ (cf. Theorem 5(c, d, e)). But $\tilde{f}_t^\infty(x) = \lim_n A_t^n\tilde{u}_{n+1}(x) \leqq A_t\tilde{u}_{t+1}(x) < \tilde{u}_t(x)$, thus contradicting the supposed $\succcurlyeq$-maximality of $\{\tilde{u}_t\}$ in U.

Finally, we show that $\tilde{u}_t = f_t$ for all t. Of course, $\tilde{u}_t \geqq f_t$, and for every $\varepsilon > 0$, there exists $\pi \in \Pi^*$ such that $H_t^n(\pi)\tilde{u}_{n+1} \geqq A_t^n\tilde{u}_{n+1} - \varepsilon$ for all t and $n \geqq t$ (cf. Lemma 2). But then $v_t(\pi) = \tilde{v}_t(\pi) = \lim_n H_t^n(\pi)\tilde{u}_{n+1} \geqq \tilde{u}_t - \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f_t = \sup_{\pi\in\Pi} v_t(\pi) \geqq \tilde{u}_t$, and, therefore, $f_t = \tilde{u}_t$. $\square$

This proof can be viewed as follows. One wishes to obtain a sequence $\{\tilde{u}_t\} \in U$ such that $\tilde{u}_t = A_t\tilde{u}_{t+1}$; then Lemma 2 will guarantee that $\tilde{u}_t = f_t$. With this objective in mind, the original problem with terminal value sequence $\{u_t\}$ is taken, and a candidate $\{f_t^\infty\}$ is obtained. Unfortunately, $\{f_t^\infty\}$ will not generally satisfy $f_t^\infty = A_t f_{t+1}^\infty$; see the example given earlier. But since $\{f_t^\infty\}$ does satisfy $f_t^\infty \geqq A_t f_{t+1}^\infty$, a new problem is considered, using $\{f_t^\infty\}$ as terminal value sequence. Fortunately, this new problem is equivalent to the original in that the infinite horizon value functions are the same. From this problem a second candidate is produced, denoted, say, by $\{f_t^{\infty 2}\}$, where $f_t^{\infty 2} \equiv \lim_n A_t^n f_{n+1}^\infty$. Continuing in this fashion, candidates $\{f_t^{\infty k}\}$ are generated, where $f_t^{\infty k} \geqq A_t f_{t+1}^{\infty k}$. Since $\{f_t^{\infty k}\}$ is decreasing in k, a new candidate $\{f_t^{\infty\omega} \equiv \lim_k f_t^{\infty k}\}$ is derived. But this is still not enough; only $f_t^{\infty\omega} \geqq A_t f_{t+1}^{\infty\omega}$ holds in general. (The reader can modify the example previously given so that this is the case.) From $\{f^{\infty\omega}\}$, though, a new candidate can be computed, $\{f_t^{\infty\omega+1} \equiv \lim_n A_t^n f_{n+1}^{\infty\omega}\}$, and so on, through all the ordinals. Taking the limit of this (decreasing) transfinite sequence yields the desired $\{\tilde{u}_t\}$. (Of course, the axiom of choice is applied in the process.)

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Besides (hopefully) motivating the proof of Corollary 8, this discussion provides the following desirable corollary.

COROLLARY 7. *If* (R), (PAε), (GDC), (DMC) *and* (N) *hold, then the greatest minorant* $\{g_t\}$ *of* $\{u_t\}$ *which satisfies* $g_t \leqq A_t g_{t+1}$ *is* $\{f_t\}$.

Proof. Given such a $\{g_t\}$, simple induction shows that $g_t \leqq A_t^n g_{n+1} \leqq A_t^n u_{n+1} = f_t^n$; thus $g_t \leqq f_t^\infty$. One then shows that $g_t \leqq f_t^{\infty 2}$ and so on; by a transfinite induction, $g_t \leqq f_t^{\infty\alpha}$ for every ordinal α, where $f_t^{\infty\alpha}$ is defined as in the previous discussion. Since $f_t = \inf_\alpha f_t^{\infty\alpha}$, we then have $g_t \leqq f_t$. $\square$

Some discussion of (GDC) seems in order. It is not satisfied by the space of Borel measurable functions on, say, [0, 1]. To see this, let Λ be a non-Borel measurable subset of [0, 1] (necessarily uncountable) and linearly order Λ (denote the order by $\succcurlyeq$) so that $(\Lambda, \succcurlyeq)$ is equivalent to the first uncountable ordinal, 2^ω. For every $\lambda \in \Lambda$, let $f_\lambda \equiv -1_{\{\lambda' : \lambda' \preccurlyeq \lambda\}}$; f_λ is measurable as $\{\lambda' \in \Lambda : \lambda' \preccurlyeq \lambda\}$ is countable. But $\inf_\lambda f_\lambda = -1_\Lambda$. Clearly, (GDC) is not satisfied by the space of continuous functions on, say, [0, 1], as not even (DC) holds for this space.

Still, (GDC) is satisfied by some important classes of functions: the (set of) convex (or concave) functions (on a convex set); any set of equi-continuous functions; the (non) increasing (or decreasing) functions on $\mathbb{R}$. Furthermore, the following result can be shown: if Ω_t is countable (for every t), then (GDC) is equivalent to (DC); see [8].

4. Equivalent terminal value sequences. The results of §§ 2 and 3 basically derive from assumptions (P*) (or (P)) and (N); the other assumptions are, in a sense, made for technical purposes. But (P*) and (N) depend critically on the terminal value sequence $\{u_t\}$. In this section, we present a method of making these analyses more robust, by relaxing (P*) and (N).

The critical notion is that of an *equivalent terminal value sequence*. Suppose a problem is given with terminal value sequence $\{u_t\}$ and associated value functions $v_t^n(\pi)$, $v_t(\pi)$, etc. Suppose also that the same problem is considered, but with different terminal value sequence $\{\tilde{u}_t\}$ and associated value functions $\tilde{v}_t^n(\pi)$. The terminal value sequences $\{u_t\}$ and $\{\tilde{u}_t\}$ are called **-equivalent* if $\tilde{v}_t(\pi) = v_t(\pi)$ for all $\pi \in \Pi^*$, and are called *equivalent* if $\tilde{v}_t(\pi) = v_t(\pi)$ for all $\pi \in \Pi$. The idea is simply that, as far as the infinite horizon values are concerned, the results are the same. With this, relaxed forms of (P*), (P) and (N) can be given.

ASSUMPTION EP* (Essentially positive*). There exists a terminal value sequence $\{\tilde{u}_t\}$ (with $\tilde{u}_t \in V_t^*$)*-equivalent to the given sequence $\{u_t\}$ such that $\tilde{u}_t \leqq H_t(\pi)\tilde{u}_{t+1}$ for all $\pi \in \Pi^*$ and all t.

An assumption (EP) is formed in a similar fashion, but the sequence $\{\tilde{u}_t\}$ must be equivalent to the given sequence and $\tilde{u}_t \leqq H_t(\pi)\tilde{u}_{t+1}$ must hold for all $\pi \in \Pi$.

ASSUMPTION EN (Essentially negative). There exists a *-equivalent sequence $\{\tilde{u}_t\}$ ($\tilde{u}_t \in V_t^*$) such that $\tilde{u}_t \geqq A_t \tilde{u}_{t+1}$.

Furthermore, the conjunction of assumptions (EP*) and (EN) is called *Assumption* C* (Convergent*) and the conjunction of (EP) and (EN) is (C). (This follows the usage of Hinderer [5].)

Corollaries to Theorems 4, 5 and 6 can easily be constructed, where (P*) is replaced by (EP*), etc. Of course, Theorem 4(f) and Corollary 7 will not hold for the original terminal value functions. Also, for problems satisfying (C*) or (C)

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

(and some of the other assumptions), corollaries can be given which combine these results. For example:

COROLLARY 8. *If* (R), (PAε), (IC), (IMC) *and* (C) *hold, then*

(a) $v_t(\pi) = H_t(\pi)v_{t+1}(\pi)$ *for all* $\pi \in \Pi$,

(b) $f_t \in V_t^*$ *and* $f_t = A_t f_{t+1}$ (*for all* t),

(c) *for every* $\varepsilon > 0$, *there exists an* ε*-optimal strategy, and*

(d) *a strategy is optimal iff it is conserving iff it is unimprovable.*

Proof. Parts (a) and (b) follow immediately from Theorem 4. Part (c) is proved exactly as Theorem 5(f) is proved (note that (DC) and (DMC*) are not needed because (a) follows from Theorem 4(c)), and (d) is the conjunction of Theorem 4(f) and Theorem 5(h,i). □

It is possible to develop conditions sufficient to guarantee that two sequences of terminal value functions are (*-) equivalent. An example is the following result, which includes Hinderer's [5] essentially positive problems as a special case. An analogous result can be given for (EN).

LEMMA 9. *If* (R) *and* (PAε) *hold, and if there exist* $\{\tilde{u}_t\}$, $\tilde{u}_t \in V_t^*$, *and sequences of* (*finite*) *nonnegative scalars* $\{w_t\}$ *and* $\{W_t\}$, *such that*

(a) $H_{t\delta}u_{t+1} \geqq u_t - w_t$ *for all* $\delta \in \Delta_t^*$ (*and all* t),

(b) $\tilde{u}_t = u_t - W_t$, *and*

(c) $W_t = \lim_n (w_t + \alpha_t w_{t+1} + \cdots + \alpha_t \alpha_{t+1} \cdots \alpha_{n-1} w_n)$,

then $\{\tilde{u}_t\}$ *and* $\{u_t\}$ *are* *-*equivalent and* (EP*) *holds.*

5. Discussion. There are two important differences between our approach here and the standard approach. The first concerns the definition of the infinite horizon problem. Here, the infinite horizon is obtained indirectly, as the limit of the finite horizon problems for the specified terminal value sequence. Thus the terminal value sequence can be critical; changing to another (nonequivalent) terminal value sequence changes the derived infinite horizon problem. But elsewhere, the infinite horizon values are frequently defined directly (however, see also [12]). This is not only natural, but essential, for expected utility problems (cf. [7]). (In these problems, it is the finite horizon problems which are defined indirectly; e.g., in the expected sum of rewards case, it is usual to define the finite horizon values as the expected partial sums, but one could conceivably change this.) Thus, in using this approach, if one has a problem where the infinite horizon values have a natural definition, one must be careful to choose the terminal value sequence so that the limit of the finite horizon values agrees with this natural definition. Also, in such cases it is sometimes easy to prove from first principles the basic equations $v_t(\pi) = H_t(\pi)v_{t+1}(\pi)$ and $f_t = A_t f_{t+1}$. Since we had to work very hard to obtain these (especially the latter in the negative case), such "first principles" proofs should be kept in mind.

The second difference between this approach and those taken in [1], [2], [3], [5], [13] lies in our assumption (PAε). For example, this assumption rules out Blackwell's [1] counterexample to the existence of ε-optimal strategies. This approach allows us to concentrate fully on issues falling outside the realm of measure theory. It is most naturally assumed in cases where the state spaces are discrete and the policy spaces are simply the cross products of the sets of feasible

actions. However, it also holds in some problems with nondiscrete state spaces (cf. Maitra [9], [10]).

REFERENCES

[1] D. BLACKWELL, *Discounted dynamic programming*, Ann. Math. Statist., 36 (1965), pp. 226–235.

[2] ———, *Positive dynamic programming*, Proc. 5th Berkeley Symposium on Mathematical Statistics, 1, University of California Press, Berkeley, 1965.

[3] D. BLACKWELL, D. FREEDMAN AND M. ORKIN, *The optimal reward operator in dynamic programming*, Ann. Probability, 2 (1974), pp. 926–941.

[4] E. DENARDO, *Contraction mappings in the theory underlying dynamic programming*, SIAM Rev., 9 (1967), pp. 165–177.

[5] K. HINDERER, *Foundations of Nonstationary Dynamic Programming with Discrete Time Parameter*, Springer-Verlag, New York, 1970.

[6] A. HORDIJK, *Dynamic Programming and Markov Potential Theory*, Mathematisch Centrum, Amsterdam, 1974.

[7] D. KREPS, *Markov decision problems with expected utility criteria*, Ph.D. dissertation, Dept. of Operations Research, Stanford University, Stanford, Calif., 1975.

[8] D. KREPS AND E. PORTEUS, *On the optimality of structured policies in countable stage decision processes. II: Positive and negative problems*, Research Paper 260, Graduate School of Business, Stanford University, Stanford, Calif., 1975.

[9] A. MAITRA, *Discounted dynamic programming on compact metric spaces*, Sankhyā, Ser. A, 30 (1968), pp. 211–216.

[10] ———, *A note on positive dynamic programming*, Ann. Math. Statist., 40 (1969), pp. 316–318.

[11] E. PORTEUS, *On the optimality of structured policies in countable stage decision processes*, Management Sci., 22 (1975), pp. 148–157.

[12] M. SCHAL, *Conditions for optimality in dynamic programming and for the limit of n-stage optimal policies to be optimal*, Z. Wahrscheinlichkeitstheorie und Verw. Gebiete, 32 (1975), pp. 179–196.

[13] R. STRAUCH, *Negative dynamic programming*, Ann. Math. Statist., 37 (1966), pp. 871–890.

Downloaded 03/04/26 to 74.72.144.134 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy